

LLMs provide a great, realized stage to imagine how our brains just might be using abstract knowledge to competently perceive, decide, and converse.¹ The key characteristic of LLMs that I want to extrapolate to the brain and mind is that of the utter apprehension of the attention mechanisms by the structures given as input to the models.

Attention, while not *all you need*, has been the insight responsible for the heralding of LLMs. Attention is where words can collaborate and relate to each other; it is where context is operationalized; it is where analogy occurs within the model. Attention patterns are functions of input tokens, embedding encoders, the Query and Key weights, and every weight in the model that comes before them. Everything mentioned here except the input tokens are trained to work well with each other. Indeed they are trained to allow for attention to occur profitably as a function of the input.

The whole idea is a model that uses context to weave plausible subsequent context. The weights serve at the pleasure of letting inputs collide into attention patterns.

Once a model is deployed for inference, it is easier to see this relationship because the weights are static. The model does not learn on the job; and if it does, the learning appears in the temporary attention patterns, again, created as a collaborative effort by all of the weights to honor the semantic contributions of every token of input.

When a model infers upon a user's query, its attention stages are just reflexively, mechanically, immediately, seamlessly apprehended into shape by the inputs. This wave of apprehension coursing through the inner states of the model quite literally becomes the output. The intelligence is in this wave of apprehension as it crests and crashes ashore with a new token; and back all of this is pulled in order to churn and apprehend itself once more.

The beauty to me is that at the very end of inference, excusing the pseudorandomness, there is one reflexive choice to be made; the context has been answered and the token is lurched out. Before the last layer of the model, there is but one more transformation to occur; it can only go one way. Before this penultimate stage, there is a checkpoint where again there is but one path to follow. We can propagate this thinking backwards through the whole model.

The point is that (for a trained model) the input is the sole determinant, the initial condition which will be unleashed into this chaotic, yes, but magnificently organized system. The input tokens simply apprehend the entire trajectory. They make it such that there is a potential energy, some unexhausted determinism at the crest of a hill, *but one path* down which the ball can roll, releasing the potential with the thrusting out of a token.

¹ I'd like to have a decorator function to apply to all that I write about LLMs, such that no matter what of mine was read, it would be prepended by a disclaimer about the sanity and non-insanity of what I mean by drawing analogical insights from LLMs for use in thinking about the brain. LLMs are so much less than brains, they are so much more than brains. They are two entirely different things. Yet they share common characteristics. That's all for today.

The inputs create a kind of gravity. They pull every weight in the model into place. The weights are trained to be so attentive! The internal state of the model during inference—and this need not be attention, but at least interpretability is not completely lost there...this can be in any circuit of weights, if you could interpret them—is a perfect reflection of the input; it has pulled itself around the input so well; the input shocks the cytoplasm of the model, folding proteins into their ordained, commanded positions. The state is utterly warped by the input. The vector field is snapped into shape. And things can only go one way—*and out pops a token*.

The teleological agent here is the semantic pattern in the inputs. It forms a structure—my choice of analogy is a molecule, with contours and regions that represent and flaunt its ornately unique charge, determining where it can and cannot bind, and so on, what conditions must be constructed around it *for* binding—which the model envelops and whose form the model takes.

Inference is a lurching, a reflex. It is the stimuli that direct the reflex potentials housed in the superposed parameters of the model. Ugh, it is magnificent to imagine, projected down, down so many dimensions into the vague 3 swimming in front of my would-be mind's would-be eyes.

The idea expressed in these words—some attempt at an incarnation of the ideas swimming in my head—comes to life as my eyes reread them, as the weights in my brain are apprehended to serve a representative pattern...Ahh, and where does that pattern exhaust itself but right back into *these* words.